

Math Summer Homework for Rising Seventh Graders

Dear almost Seventh Grader!

I'm looking forward to teaching your math class in the fall. Seventh grade math is the bridge between pre-algebra and algebra. We'll develop our understanding of probability (the likelihood of something happening) and geometry (the characteristics and measurements of shapes), and we'll use our knowledge of patterns to find the values of unknowns in problems using variables! We'll use numbers, diagrams, tables, and graphs to help us solve and understand problems.

In math class, we'll be working to ensure we understand what a solution means. You'll be explaining your thinking to your group and your class often and will have lots of opportunities to improve your communication throughout the year. I'm looking forward to helping all of you succeed and grow!

Some skills that will be important to review and keep fresh over the summer are fraction and negative number operations (addition, subtraction, multiplication, and division); calculating the area and perimeter of rectangles, triangles, and circles; and solving problems. Please do as much as you can in this packet. **If you are unsure of how to solve a problem, write down your thoughts about it or try to at least start the problem. Circle any problems that are hard for you so that I can get an idea of what we need to review together back at school.**

It is best if you complete a couple of sheets from this math packet each week over the summer. That should be about 30-40 minutes every week. If you leave the packet until the end of the summer, you will feel like there is too much to do and may feel overwhelmed. If you sprinkle it throughout your summer, you'll keep your math brain fresh and hardly notice that you have math homework! **You will be turning this packet in on the first day of school.**

I'm looking forward to getting to know you all soon!

Have a great summer!

--Edward Pertcheck

Four 4s

For the First Week of School

We'll use your work on this page for an activity in class the first week of school!

Using **only and exactly four 4s** and **any mathematical operation**, create expressions equaling as many of the numbers from 0-20 as possible. If you can find multiple ways to make each number, record multiple ways in the boxes below.

Some notes: exponents and square roots count as operations

For example: $4 \times 4 - 4 \times \sqrt{4}$
 $= 16 - 4 \times 2$
 $= 8$

$4^2 - 4 - 4 - 4$
 $= 16 - 12$
 $= 4$

<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>
<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>

Four 4s

For the First Week of School

We'll use your work on this page for an activity in class the first week of school!

<u>9</u>	<u>10</u>	<u>11</u>	<u>12</u>
<u>13</u>	<u>14</u>	<u>15</u>	<u>16</u>
<u>17</u>	<u>18</u>	<u>19</u>	<u>20</u>

Candy Dilemma

I bought a box of candy for myself last week. However, by the time I got home I had eaten $\frac{1}{4}$ of the candies. As I was putting the groceries away, I ate $\frac{1}{2}$ of what was left. There are now 6 chocolates left in the box.

How many chocolates were in the box to begin with?

Be sure to show and explain all of your reasoning.



Molly Mathematician's Locker Combination Predicament

Molly Mathematician can't remember her gym locker combination. The Physical Education teacher won't provide her with the combination, but did give her the following clues about the combination:

1. The 3 numbers in the combination are between the numbers 15 and 30.
2. The sum of the numbers is 70.
3. When you square each number in her locker combination the digits 1 – 9 will appear only once in the digits of the 3 square numbers.

What is the combination to Molly's locker?

Show all of your work and tell how you achieved a solution by labeling all that you do.

Sink or Swim?

In New Zealand there is a river known as the Kaituna. On it, there exists a 23-foot waterfall; some people like to go over the waterfall on a raft. A group of 6 people decide to give it a try; they consist of four adults and two teenagers.

The raft **MUST** be traveling at least 30 miles per hour in order to successfully go over the falls; otherwise it will tip over. The river flows at one-sixth of the needed speed. An adult can paddle the raft at a speed of 5 miles per hour. Three teenagers can paddle the same speed as two adults.

How fast is the raft traveling?

Will the group be able to make it over the falls without tipping over?



The Triangle Train

Find the perimeter of a train of 100 equilateral triangles if the triangles are joined side to side.

Each side is one inch long.

Write an equation to determine the perimeter of the triangle train.

Mr. Larson is planning the seating for a school recital. He needs to reserve $\frac{1}{3}$ of the seats for students and $\frac{1}{6}$ of the seats for parents.

- a. After reserving seats for students and parents, what fraction of the seats in the auditorium are left?
- b. Mr. Larson's principal tells him that he also needs to reserve $\frac{1}{8}$ of the seats for teachers and school officials. The remainder can be used for open seating. What fraction of the seats are now left for open seating?
- c. Later, Mr. Larson's principal says he should reserve $\frac{1}{4}$ of the seats for students from other middle schools. Are there enough seats left? If not, explain why not; otherwise, state what fraction of the seats will be available for open seating.

Multiplying and Dividing Fractions

Three groups of students are sharing leftover pizza (all the same size originally). In which group does each student get the most pizza? Explain your choice.

A. Six students equally share $\frac{3}{4}$ of a pizza.

B. Three students equally share $\frac{1}{3}$ of a pizza.

C. Four students equally share $\frac{2}{3}$ of a pizza.

Find each quotient.

2. $\frac{1}{12} \div \frac{5}{6}$

3. $4 \div \frac{1}{3}$

4. $6 \div \frac{3}{4}$

5. $5 \div \frac{9}{10}$

6. $8 \div \frac{2}{3}$

7. $\frac{4}{5} \div 2$

8. $\frac{7}{8} \div 3$

9. $\frac{5}{6} \div 5$

10. $\frac{4}{9} \div 8$

Find Each Product

11. $\frac{3}{5}$ of $\frac{3}{4}$

12. $\frac{1}{2} \times \frac{1}{3}$

13. $\frac{1}{8} \times \frac{3}{4}$

14. $\frac{2}{5} \times \frac{7}{11}$

15. $\frac{2}{3}$ of $\frac{1}{4}$

16. $\frac{2}{5} \times \frac{1}{2}$

17. $\frac{1}{4}$ of $\frac{4}{5}$

18. $\frac{5}{6} \times \frac{2}{5}$

Adding and Subtracting Fractions

Find each sum or difference.

1. $\frac{1}{4} + \frac{2}{4}$

2. $\frac{7}{10} - \frac{4}{10}$

3. $\frac{5}{8} - \frac{3}{8}$

4. $\frac{1}{8} + \frac{5}{8}$

5. $\frac{5}{8} + \frac{2}{8}$

6. $\frac{3}{10} + \frac{6}{10}$

7. $\frac{2}{5} - \frac{1}{10}$

8. $\frac{5}{8} - \frac{1}{4}$

9. $\frac{3}{10} + \frac{4}{5}$

10. $\frac{11}{16} + \frac{5}{8}$

11. $\frac{2}{3} - \frac{1}{6}$

12. $\frac{3}{5} + \frac{7}{10}$

13. What is the total amount of sugar in the recipe at the right?

14. Martha decides to double the recipe. How much brown sugar will she use?

Martha's Cookie Recipe

1 cup shortening

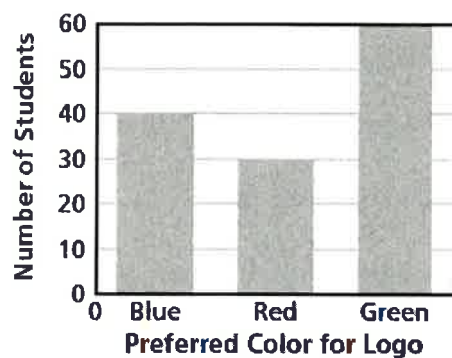
2 eggs

 $\frac{1}{4}$ cup white sugar $\frac{1}{4}$ cup brown sugar $1\frac{1}{2}$ cups flour

1 teaspoon vanilla

Percents

1. The student council at Metropolis Middle School conducted a survey to see whether students would prefer blue, red, or green as the new color for the school logo. The results of the survey are shown in the bar graph below.



- What is the total number of students who were surveyed?
 - What percent of students surveyed preferred blue?
 - What percent of students surveyed preferred red?
 - What percent of students surveyed preferred green?
 - If 970 students attend Metropolis Middle School, what percent of the students were surveyed?
2. What percent of 75 is 40?
3. What percent of 45 is 135? Explain your reasoning.

Probability Darts

Find the portion of the dart board that each panel occupies and use your knowledge of degrees and fractions to answer the following questions about probability.

REMEMBER: Probability is the likelihood a given outcome will occur. It is expressed as a fraction.

Fractions

A $\frac{1}{18}$

$20^\circ/360^\circ$

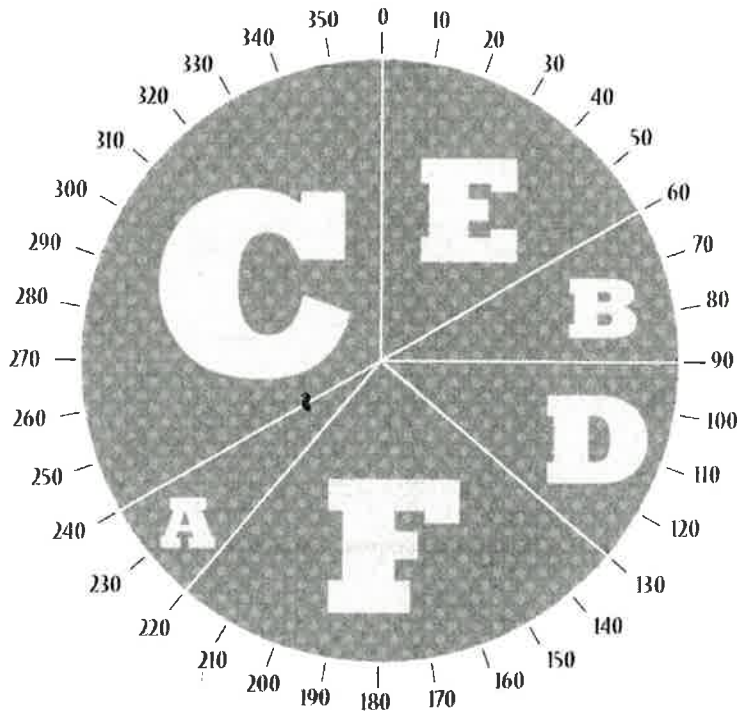
B _____

C _____

D _____

E _____

F _____



Use the information above to answer the questions below.

- 1 Is the next dart more likely to hit panel C, E, or F? Why?
- 2 What is the probability that the next dart thrown hits a panel with a letter that comes after C alphabetically?
- 3 Write the letters of the panel in the order of most to least probable of being hit.

Review: Fractions

Add or subtract.

Choose 4 rows to complete in this section

A. $\frac{5}{6} + \frac{5}{8} =$ _____	$\frac{13}{21} + \frac{5}{7} =$ _____	$\frac{5}{6} + \frac{10}{18} =$ _____	$\frac{6}{7} + \frac{8}{9} =$ _____
B. $\frac{4}{5} + \frac{9}{15} =$ _____	$\frac{5}{16} + \frac{17}{8} =$ _____	$\frac{13}{12} + \frac{7}{8} =$ _____	$\frac{14}{24} + \frac{7}{12} =$ _____
C. $\frac{27}{30} - \frac{5}{6} =$ _____	$\frac{5}{6} - \frac{1}{5} =$ _____	$\frac{7}{8} - \frac{1}{2} =$ _____	$\frac{5}{6} - \frac{2}{9} =$ _____
D. $\frac{9}{16} - \frac{3}{8} =$ _____	$\frac{4}{5} - \frac{2}{8} =$ _____	$\frac{4}{7} - \frac{3}{14} =$ _____	$\frac{3}{4} - \frac{1}{5} =$ _____
E. $\frac{3}{6} - \frac{2}{15} =$ _____	$\frac{5}{8} - \frac{1}{6} =$ _____	$\frac{7}{9} - \frac{2}{6} =$ _____	$\frac{7}{24} - \frac{3}{12} =$ _____
F. $2\frac{1}{8} + 4\frac{1}{2} =$ _____	$3\frac{1}{2} + 5\frac{3}{6} =$ _____	$3\frac{9}{10} + 2\frac{4}{15} =$ _____	$2\frac{2}{3} + 3\frac{1}{6} =$ _____
G. $3\frac{7}{9} + 2\frac{3}{27} =$ _____	$4\frac{2}{8} + 3\frac{4}{16} =$ _____	$3\frac{2}{12} + 3\frac{1}{3} =$ _____	$6\frac{4}{9} + 2\frac{2}{3} =$ _____
H. $6\frac{7}{8} + \frac{2}{6} =$ _____	$6 + 3\frac{5}{9} =$ _____	$\frac{3}{12} + 5\frac{9}{12} =$ _____	$12\frac{4}{8} + \frac{2}{4} =$ _____

Compare. Use >, <, or =.

Choose 2 rows to complete in this section

I. $5\frac{1}{4} - 1\frac{1}{8} \square 5\frac{4}{6} - 1\frac{1}{3}$	$6\frac{5}{18} + 1\frac{3}{9} \square 3\frac{1}{4} + 3\frac{4}{6}$	$\frac{6}{12} \square \frac{9}{24}$
J. $7\frac{1}{2} - 4 \square 9 - 7\frac{4}{10}$	$8\frac{7}{9} - 4\frac{1}{3} \square 9\frac{5}{6} + 5\frac{2}{3}$	$\frac{10}{21} \square \frac{5}{7}$
K. $9 + 3\frac{4}{5} \square 15 - 4\frac{2}{3}$	$9\frac{4}{10} + 2\frac{3}{5} \square 10 + 3\frac{4}{9}$	$\frac{5}{15} \square \frac{2}{3}$
L. $7\frac{1}{4} - 2\frac{2}{8} \square 3\frac{5}{8} - 1\frac{1}{3}$	$4\frac{7}{18} + 1\frac{3}{9} \square 2\frac{1}{2} + 3\frac{1}{2}$	$\frac{3}{12} \square \frac{4}{8}$

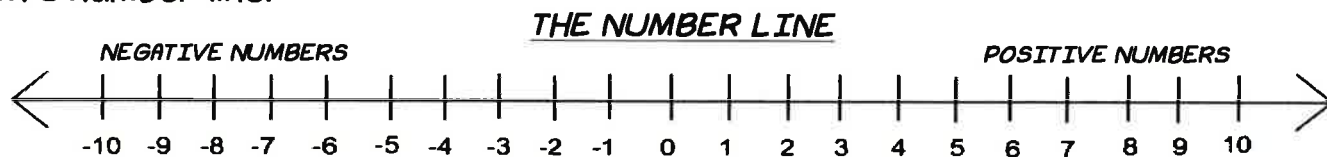
Write the missing number.

Complete both rows in this section

M. $5\frac{2}{9} + \underline{\hspace{2cm}} = 11$	$\underline{\hspace{2cm}} - 3\frac{2}{7} = 7\frac{5}{21}$	$\underline{\hspace{2cm}} + 2\frac{2}{3} = 5\frac{5}{6}$
N. $\underline{\hspace{2cm}} - 6\frac{3}{5} = 3\frac{1}{3}$	$\underline{\hspace{2cm}} - 7\frac{1}{6} = 4\frac{2}{3}$	$\underline{\hspace{2cm}} - 5\frac{3}{4} = 9\frac{5}{8}$

INTEGER CHEAT SHEET

Integers- A set of positive and negative whole numbers. They can be represented on a number line.



Absolute Value- The distance a number is from zero on the number line. An absolute value is never negative. Examples: $|-5| = 5$ and $|5| = 5$

ADDING INTEGERS

SAME SIGN- Add and Keep the Sign!

Add the absolute value of the numbers and keep the same sign.

$$(\text{positive}) + (\text{positive}) = \text{Positive}$$

$$(+4) + (+5) = +9$$

$$(\text{negative}) + (\text{negative}) = \text{Negative}$$

$$(-4) + (-5) = -9$$

DIFFERENT SIGNS- Subtract and Keep the Sign of the Bigger Number!

Subtract the absolute value of the numbers and keep the sign of the bigger number.

$$(-4) + (+5) = +1$$

$$(+4) + (-5) = -1$$

SUBTRACTING INTEGERS

Do not subtract integers. You must change the signs:

"Add the Opposite"

KEEP- Keep the sign of the first number

CHANGE- Change the subtraction sign to addition

CHANGE- Change the sign of the second number to the opposite sign. If it is positive- change to negative. If it is negative- change to positive.

$$(+4) - (-4)$$

$$\begin{array}{ccc} \text{Keep} & \text{change} & \text{change} \\ (+4) & + & (+4) \end{array}$$

NOW USE THE RULES FOR ADDING:
SAME SIGN- Add absolute values and keep sign:

$$(+4) + (+4) = 8$$

MULTPLYING INTEGERS

SAME SIGNS- POSITIVE

Multiply the numbers. Answer will be positive.

$$(-5) \times (-5) = +25$$

DIFFERENT SIGNS- NEGATIVE

Multiply the numbers. Answer will be negative

$$(+5) \times (-5) = -25$$

DIVIDING INTEGERS

SAME SIGNS- POSITIVE

Divide the numbers. Answer will be positive.

$$(-5) \div (-5) = +1$$

DIFFERENT SIGNS- NEGATIVE

Divide the numbers. Answer will be negative

$$(+5) \div (-5) = -1$$

All Operations with Integers (A)

Use an integer strategy to find each answer.

$-8 - -9 =$

$12 - -2 =$

$50 \div -5 =$

$-4 - 11 =$

$1 - 11 =$

$8 \times -1 =$

$9 + 3 =$

$5 \times -5 =$

$2 - 4 =$

$-9 - 6 =$

$4 \times 3 =$

$2 + 7 =$

$30 \div -6 =$

$-2 + -1 =$

$-6 + -11 =$

$5 + 9 =$

$7 + -12 =$

$-11 + -3 =$

$-12 - 10 =$

$11 - 7 =$

$5 \times 7 =$

$48 \div -12 =$

$-9 + 10 =$

$-9 + 2 =$

$9 \times 6 =$

$-5 - -1 =$

$6 \times -12 =$

$100 \div 10 =$

$12 \div -4 =$

$8 \times 4 =$

Basic Algebra Tables

Complete the tables. Write a full equation in each empty box.

	$18 - x$	$3x$	$\frac{40}{x}$
$x = 2$	$18 - 2 = 16$		
$x = 5$		$3 \times 5 = 15$	
$x = 8$			$40 \div 8 = 5$
$x = 10$	$18 - 10 = 8$		

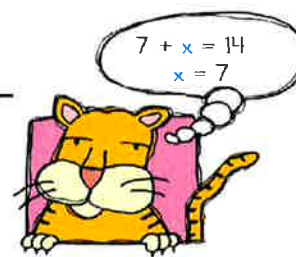
	$9y$	$\frac{y}{3}$	$y + y$
$y = 3$			
$y = 6$			
$y = 9$			
$y = 12$			

Complete the table. Write equations in the empty white boxes. Also, write three algebraic expressions in the gray boxes along the top.

$z = 1$	$1 \times 3 = 3$		$28 \div 1 = 28$
$z = 4$	$4 \times 3 = 12$	$4 + 8 = 12$	
$z = 7$		$7 + 8 = 12$	

Name: _____

Algebraic Expressions



Tell whether each algebraic equation is correct.
Write true or not true on the line next to each.

1. $a - 6 = 4, a = 10$ _____
2. $\frac{c}{12} = 2, c = 6$ _____
3. $\frac{d}{6} = 2, d = 18$ _____
4. $8z = 48, z = 7$ _____
5. $x + x + 7 = 16, x = 4$ _____
6. $\frac{30}{5} = h, h = 6$ _____
7. $\frac{8}{i} = 1, i = 8$ _____
8. $13 - z = 8, z = 4$ _____

If the algebraic expression shown is true, write true on the line. If the algebraic expression is not true, cross out the value for the variable and write a new value on the line to make it true.

example: $a - 1 = 6, a = 8$ _____

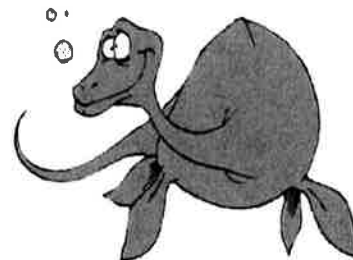
Since this is not true, cross out the 8 and write a = 7 on the line.

9. $9 - w = 1, w = 9$ _____
10. $\frac{p}{2} = 5, c = 10$ _____
11. $\frac{18}{k} = 2, k = 9$ _____
12. $12 + j = 21, j = 6$ _____
13. $7a = 28, a = 3$ _____
14. $\frac{24}{y} = 4, y = 8$ _____
15. $\frac{36}{6} = b, b = 3$ _____
16. $17 - v = 5, v = 12$ _____

17. Is this equation in the box to the right always false, no matter what value you give the variable? Explain your answer.

$0b = 4$

Determine the value of the variable in each equation.



1. $6 + a = 12$

$a = \underline{\hspace{2cm}}$

2. $7 - b = 2$

$b = \underline{\hspace{2cm}}$

3. $11 + 14 = c$

$c = \underline{\hspace{2cm}}$

4. $\frac{24}{d} = 3$

$d = \underline{\hspace{2cm}}$

5. $10e = 110$

$e = \underline{\hspace{2cm}}$

6. $\frac{f}{7} = 7$

$f = \underline{\hspace{2cm}}$

7. $13g = 26$

$g = \underline{\hspace{2cm}}$

8. $35 - h = 10$

$h = \underline{\hspace{2cm}}$

9. $6 + i = 23$

$i = \underline{\hspace{2cm}}$

10. $j - 17 = 7$

$j = \underline{\hspace{2cm}}$

11. $\frac{42}{7} = k$

$k = \underline{\hspace{2cm}}$

12. $4m = 32$

$m = \underline{\hspace{2cm}}$

13. $\frac{72}{n} = 9$

$n = \underline{\hspace{2cm}}$

14. $33 + 66 = p$

$p = \underline{\hspace{2cm}}$

15. $\frac{q}{8} = 5$

$q = \underline{\hspace{2cm}}$

★ $5 + r = 14 - 3$

$r = \underline{\hspace{2cm}}$

★ $11 + 4 = 3s$

$s = \underline{\hspace{2cm}}$

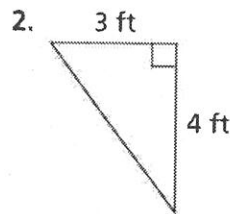
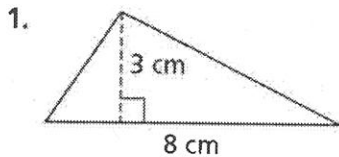
Skill: Area of Triangles

$$A = \frac{1}{2} \cdot \text{base length} \cdot \text{height}$$

(17)

Covering

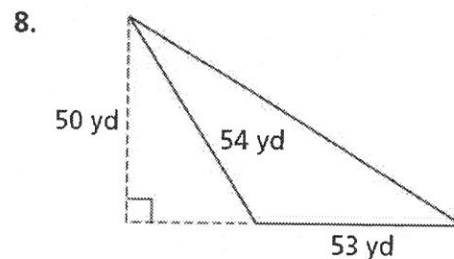
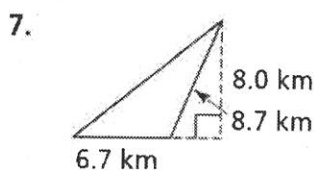
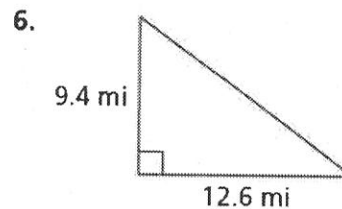
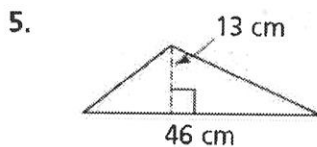
Find the area of each triangle.



Tell whether each statement is *true* or *false*.

3. Two triangles that have the same base always have the same area.
4. Any obtuse triangle has a greater area than any acute triangle.

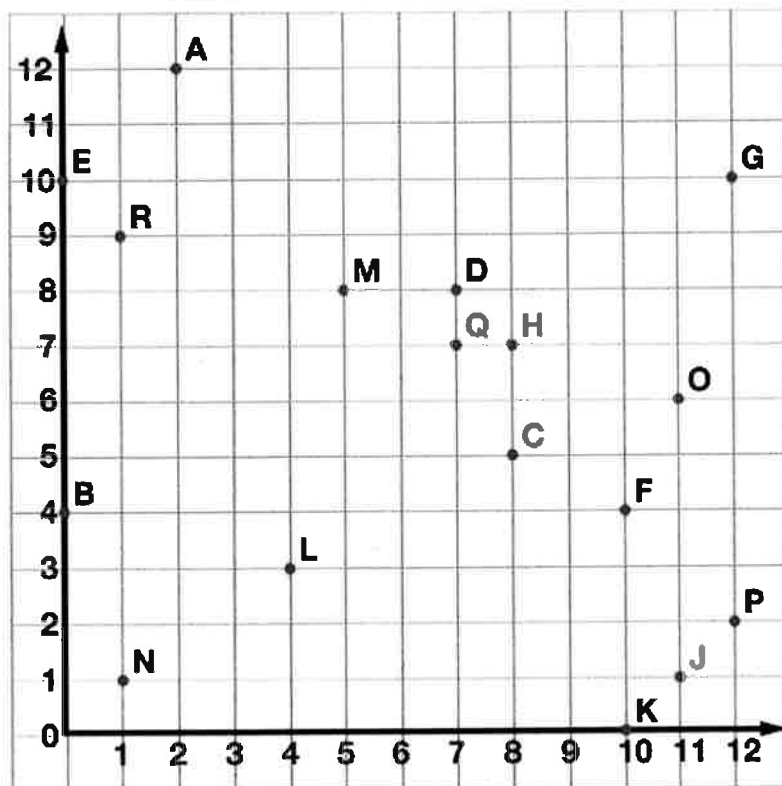
Find the area of each triangle.



Solve.

9. The area of a triangle is 6 square units. Both the height and the length of the base are whole numbers. What are the possible lengths and heights?

Ordered Pairs



Tell what point is located at each ordered pair.

- | | | |
|---------------------|--------------------|--------------------|
| 1. $(5, 8)$ _____ | 2. $(12, 2)$ _____ | 3. $(8, 7)$ _____ |
| 4. $(12, 10)$ _____ | 5. $(7, 7)$ _____ | 6. $(0, 10)$ _____ |

Write the ordered pair for each given point.

- | | | |
|-------------|-------------|-------------|
| 7. N _____ | 8. L _____ | 9. J _____ |
| 10. A _____ | 11. B _____ | 12. E _____ |

Plot the following points on the coordinate grid.

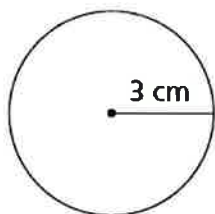
- | | | |
|-----------------|----------------|-----------------|
| 13. S $(6, 11)$ | 14. T $(3, 5)$ | 15. U $(9, 12)$ |
|-----------------|----------------|-----------------|

Skill: Area of Circles

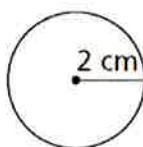
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Find the area of each circle. Round to the nearest tenth.

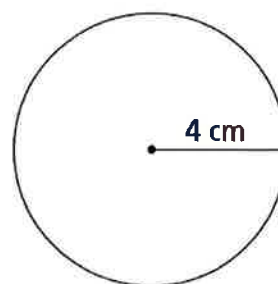
1.



2.

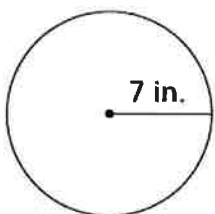


3.

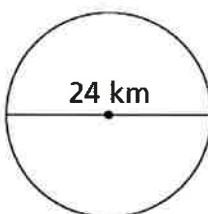


Find the area of each circle. Round to the nearest unit. Use $\frac{22}{7}$ for π .

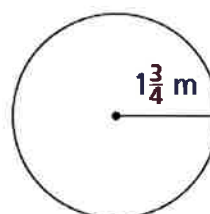
4.



5.

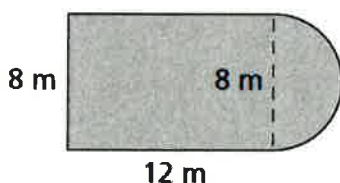


6.

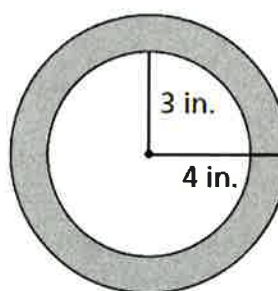


Find the area of each shaded region to the nearest tenth.

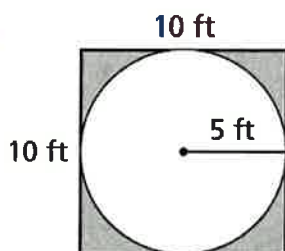
7.



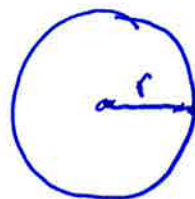
8.



9.



Some helpful information...



r = radius (the distance from the center of the circle to any point on the circumference)

$$\text{Area of a Circle} = \pi \cdot r^2$$

or

$$\text{Area} = \pi \cdot r \cdot r$$

$$\pi = 3.14...$$

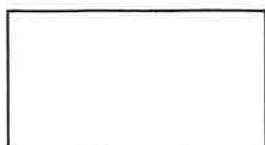
Choose half of the problems below to practice

20

Find the perimeter and area of each rectangle.

1.

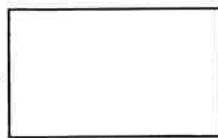
8 cm



15 cm

2.

12 in.

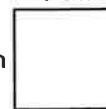


20 in.

3.

6 cm

6 cm



4. $\ell = 5$ in., $w = 13$ in.

5. $\ell = 18$ m, $w = 12$ m

6. $\ell = 3$ ft, $w = 8$ ft

7. rectangle: $\ell = 16$ mm, $w = 12$ mm

8. rectangle: $\ell = 65$ mi, $w = 48$ mi

9. The length of a rectangle is 8 centimeters. The width is 6 centimeters.

a. What is the area?

b. What is the perimeter?

10. The area of a rectangle is 45 square inches.

One dimension is 5 inches. What is the perimeter?